Binary Addition Project with Arduino

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Introduction

- Reading the Arduino introduction documentation
- Creating the GitHub repository
- Adding the base code provided by the professor
- Commenting the code in English
- Implementing full adder logic

The project started with reading the Arduino documentation, followed by creating a GitHub repository. The base code was added and commented in English to improve clarity. The full adder logic was implemented using two main functions: somaBit and somaCarryBit.

Software Demo

- Digital inputs: pins 0 to 7
- Control button: pin 13
- Visual outputs: LEDs on pins 8 to 12
- Addition of two 4-bit binary numbers

The system reads two 4-bit binary numbers through switches connected to pins 0 to 7. The button on pin 13 activates the addition operation. The result is displayed on LEDs connected to pins 8 to 12, representing the sum bits and the carry bit.

Survey of what was developed

- Functional code with full adder logic
- Explanatory comments in English
- Public GitHub repository
- PDF presentation

I developed a functional system that performs binary addition with carry. The code was commented in English and is available on GitHub. The presentation was structured to demonstrate all stages of the project.

Future development activities

- Implement binary subtraction
- Add 7-segment display
- Create graphical interface using Processing

I plan to implement binary subtraction, add a 7-segment display for better visualization, and create a graphical interface using Processing to make the project more interactive.

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Thank you for your attention!